

TARBES LOURDES PYRENEES AP, France

WMO: 076210

Lat: 43.1881N

Lon: 0.0000E

Elev: 384

StdP: 96.80

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.8	-7.3	2.1	-1.8	-5.6	2.5	-0.5	11.2	9.1	9.8	9.0	2.1	180	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	31.7	21.0	29.6	20.5	27.8	19.8	22.6	28.9	21.6	27.5	20.6	26.1	2.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	16.0	25.9	19.6	15.1	24.4	18.8	14.2	23.1	68.5	29.0	64.6	27.7	61.1	26.2	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	7.5	6.4	-7.3	35.5	1.9	1.8	-8.6	36.8	-9.8	37.8	-10.8	38.8	-12.2	40.1		
			-7.8	24.9	1.9	0.9	-9.1	25.6	-10.3	26.1	-11.4	26.5	-12.7	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	5.8	6.4	9.2	11.0	14.5	18.1	19.9	20.0	17.2	14.3	8.9	6.5	
	DBStd	6.16	3.53	3.77	3.42	3.22	3.60	3.56	3.09	3.08	3.00	3.40	3.47	3.63	
	HDD10.0	512	135	108	56	24	6	0	0	0	0	7	61	115	
	HDD18.3	2276	389	334	284	219	127	47	19	17	57	131	284	366	
	CDD10.0	1493	4	7	30	56	146	244	306	311	216	139	26	8	
	CDD18.3	214	0	0	0	1	8	41	67	69	22	5	0	0	
	CDH23.3	2022	0	1	2	17	119	429	572	593	226	63	1	0	
	CDH26.7	638	0	0	0	1	24	146	189	219	49	9	0	0	
Wind	WSAvg	2.7	2.9	3.1	3.1	3.1	2.9	2.7	2.6	2.4	2.4	2.3	2.7	2.8	
Precipitation	PrecAvg	1139	107	90	86	116	116	86	77	68	77	88	132	96	
	PrecMax	1619	263	223	188	261	208	218	147	123	172	287	282	196	
	PrecMin	916	10	19	12	44	52	23	27	15	27	35	24	11	
	PrecStd	164	64	56	41	46	39	42	31	28	41	49	61	50	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	21.0	23.0	25.7	29.6	33.6	33.9	35.0	30.3	27.9	21.5	18.5	
		MCWB	11.3	11.7	13.0	15.8	19.0	21.8	22.5	21.8	19.6	17.5	14.2	11.6	
	2%	DB	15.2	17.3	20.5	22.7	26.4	30.3	30.8	31.7	28.1	25.0	19.1	16.2	
		MCWB	10.0	10.1	12.6	14.4	17.8	20.3	21.3	21.3	18.9	16.9	13.0	10.7	
	5%	DB	13.1	15.1	18.2	20.5	24.1	28.0	28.7	29.0	26.1	22.9	16.7	14.1	
		MCWB	9.3	9.4	11.5	13.4	16.9	19.5	20.7	21.1	18.4	16.1	11.9	9.7	
	10%	DB	11.4	12.6	16.0	18.0	21.9	25.6	26.6	26.6	23.9	20.8	14.3	12.2	
		MCWB	8.8	8.6	10.5	12.4	15.8	18.8	20.0	20.2	17.5	15.5	11.1	9.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	12.4	14.6	16.7	20.3	23.6	24.1	24.0	21.7	19.1	15.6	13.0
			MCDB	15.4	17.7	20.2	23.4	26.7	29.9	30.8	30.6	27.3	24.3	19.7	16.4
2%		WB	11.2	11.2	13.1	15.3	18.7	21.7	22.3	22.6	20.2	17.8	13.8	11.7	
		MCDB	13.7	15.4	18.4	21.3	24.7	28.2	28.6	28.8	25.9	22.7	17.3	14.5	
5%		WB	10.1	10.1	12.0	14.0	17.5	20.4	21.2	21.6	19.1	16.9	12.7	10.6	
		MCDB	12.4	13.5	17.1	19.3	23.1	26.3	27.1	27.3	24.2	21.4	15.5	13.2	
10%		WB	8.9	9.2	11.0	12.9	16.3	19.2	20.3	20.6	18.1	15.9	11.6	9.5	
		MCDB	11.0	12.2	15.3	17.1	20.9	24.6	25.5	25.6	22.4	19.7	13.9	11.9	
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.5	10.5	10.3	10.2	10.5	10.2	10.4	10.8	10.6	9.1	9.3	
		MCDBR	11.9	14.4	14.8	15.1	14.6	15.5	15.2	15.3	14.6	14.1	12.9	12.4	
		MCWBR	7.8	8.6	8.3	8.2	7.7	7.4	7.1	7.1	7.0	7.5	7.8	7.9	
	5% WB	MCDBR	10.3	11.8	12.6	13.3	13.1	13.2	12.9	13.0	12.7	12.2	10.8	10.3	
		MCWBR	7.4	7.8	7.6	7.6	7.2	6.9	6.7	6.6	6.6	6.9	7.2	7.2	
Clear-Sky Solar Irradiance	taub	0.299	0.315	0.353	0.377	0.389	0.402	0.393	0.394	0.364	0.344	0.315	0.297		
	taud	2.502	2.447	2.363	2.312	2.321	2.318	2.347	2.352	2.425	2.484	2.508	2.512		
	Ebn at Noon	844	886	886	889	884	870	874	862	864	839	815	812		
	Edh at Noon	70	88	108	123	125	126	121	117	101	84	70	64		
All-Sky Solar Radiation	RadAvg	1.62	2.52	3.71	4.66	5.39	6.10	6.16	5.47	4.42	2.95	1.85	1.48		
	RadStd	0.15	0.29	0.40	0.48	0.43	0.39	0.36	0.33	0.21	0.25	0.21	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page